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Behavior and Plasma Cortisol Following Brief Peer Separation in Juvenile Squirrel Monkeys

MICHAEL B. HENNESSY[1], SALLY P. MENDOZA[2], AND JOEL N. KAPLAN[1]
[1]*Developmental Psychobiology Program, SRI International, Menlo Park, California and* [2]*California Primate Research Center, University of California, Davis*

The effects of 30-min separation from peers on behavior and plasma cortisol levels were studied in juvenile, mother-reared, female squirrel monkeys, housed in triads. Juveniles vocalized much more frequently when isolated in a novel room than they did when either undisturbed, handled only, placed with both peers in a novel room, or separated from one peer and housed with the other in the home cage. When separated from one peer in the home cage, the monkeys engaged in putative scent-marking behaviors more frequently than when undisturbed. Exposure to the novel room resulted in increased vocalizing, and decreased locomotor activity and scent marking, regardless of whether the subject was isolated or with peers. Plasma cortisol levels were elevated following all manipulations and no effects could be attributed specifically to separation or exposure to novelty. These results demonstrate that brief separation from peers can evoke a behavioral reaction in juvenile squirrel monkeys that resembles the behavioral "protest" response of maternally-separated infants, and that the response of juveniles, like that of infants, varies greatly with the separation procedure employed.

Key words: peer separation, juveniles, protest behavior, cortisol, squirrel monkey

INTRODUCTION

In the study of primate social relationships, the strength of a relationship often is assessed by examining the severity of disturbance that is caused by its disruption. This technique has most often been used to investigate the bond between mother and infant macaques [Mineka & Suomi, 1978]. Following maternal separation, infants show an initial stage of behavioral activation or "protest" characterized by increases in such behaviors as vocalizations and locomotor activity [Jensen & Tolman, 1962; Kaplan, 1970]. In some macaque species, protest is followed in several days by a "despair" reaction distinguished by inactivity and social withdrawal [Seay et al., 1962; Kaufman & Rosenblum, 1967]. These behavioral stages are paralleled by physiological changes including alterations in heart rate, body temperature, and sleep patterns [Reite & Short, 1978; Reite et al., 1978a]. Infant macaques reared on maternal surrogates exhibit behavioral and physiological disturbance following separation from the surrogate, although the ef-

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Address reprint requests to Michael B. Hennessy, PhD, 10056, SRI International, 333 Ravenswood Avenue, Menlo Park, CA 94025.

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fects are less severe. This suggests that the attachment of the infant macaque to its surrogate is less intense than is that of infant to mother [Reite et al., 1978b; Meyer et al., 1975].

Young rhesus macaques raised in peer groups without maternal contact display protest and despair following peer separation similar to that seen following separation from the mother [Suomi et al., 1970, 1976]. On the other hand, in juvenile peer groups formed following weaning from the mother, peer separation has been found to induce a protest reaction, but no apparent signs of despair [Bowden & McKinney, 1972; Erwin et al., 1971, 1973; McKinney et al., 1972]. These data suggest that young rhesus form relationships with peers that are not as strong as the bond typically formed with the mother except under conditions in which peer relationships have been used to compensate for maternal absence.

In the squirrel monkey, the subject of the present investigation, a number of studies have examined the effect of brief periods of separation on both behavior and the activity of the pituitary-adrenal system, a neuroendocrine system known to increase hormonal secretion in a wide range of arousing or stressful conditions [Hennessy & Levine, 1979]. Infant squirrel monkeys separated from their mothers show clear behavioral protest [Kaplan, 1970], unless a familiar, substitute attachment figure (i.e., maternally behaving adult female or "aunt") is available during the separation period [Rosenblum, 1971]. However, infants exhibit large increases in plasma cortisol levels 30 min following maternal separation regardless of whether the infant is left in its home cage with an aunt or placed alone in a novel room during the period of maternal absence [Coe et al., 1978]. Thus, pituitary-adrenal responses to separation can occur when changes in behavior are not apparent. In contrast to the response of mother-reared infants, surrogate-reared infants show no change in cortisol levels 30 min following removal of the surrogate from the home cage [Hennessy et al., 1979; Hennessy & Kaplan, in press]. If the infant is isolated in a novel room during separation, an elevation in cortisol levels is observed at 30 min, but this response can be accounted for by exposure to novelty since an equivalent cortisol elevation at this time point is observed if the infant is placed with the surrogate in the novel room [Hennessy & Kaplan, in press]. Overall, these findings suggest that factors attendant to the separation procedure (i.e., the availability of a substitute attachment figure and exposure to novelty) can substantially modify both behavioral and pituitary-adrenal responsiveness.

In the present study, we used the separation procedures developed in our studies of squirrel monkey infants to assess peer relationships in older animals. Specifically, the study was designed to determine if brief separation from peers would affect either the behavior or pituitary-adrenal activity of juvenile squirrel monkeys. We examined peer separation conditions in which: (1) the separated monkey had another social partner available and was left in familiar surroundings; and (2) the separated monkey was isolated from all peers in a novel room. We also independently evaluated the effects of exposure to novelty in monkeys not separated from peers. We studied females that had been reared by their biological mothers. Females appeared to be best suited for an initial investigation of this type because evidence suggests that there is greater social attraction among female squirrel monkeys than there is among males [Mason, 1971, 1974].

METHOD

Subjects and Housing

Twelve female mother-reared squirrel monkeys (*Saimiri sciureus*) of the Colombian and Peruvian varieties, that had been born and reared in the SRI colony, were used. Six animals were 20–22 months of age and six were 33–35 months of age at the start of the experiment. Squirrel monkey females typically begin attaining sexual maturity at about 2.5–3 years [Kaplan et al., 1981]. Each of the monkeys was initially housed in semi-

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permanent multianimal groups comprised primarily of juveniles. Three weeks before the beginning of the experiment, the animals were rehoused in groups of three, with monkeys of a similar age comprising each triad. Each individual triad was formed from animals that had been maintained until then in the same large group. The triads were housed in separate wire mesh cages (1.5 × 0.9 × 0.7 m) that had three metal perches positioned at different heights, extending the width of each cage. The subjects were the only animals maintained in the room. A 12-hr light/dark schedule (lights on at 0700) was imposed. The animals were fed Purina New World monkey chow and water. Vitamin-supplemented, orange-flavored drink was provided once a week.

All monkeys were tested in five conditions. In the (1) BASE condition, the monkeys were undisturbed prior to behavior observation. This condition was used to estimate baseline behavior and resting plasma cortisol levels. In the (2) handle (HAN) condition, the three monkeys of the group were caught and held separately in front of the home cage for ~30 sec, then released back into the cage, and behavior observations began. This condition provided a control for the handling required in conditions 3–5. The (3) separation-partner-home (SEP–HOM) condition was identical to HAN, except that following capture, only two of the three monkeys were returned to the home cage and observed. Thus in SEP–HOM, subjects experienced separation from a peer, but remained with another social partner and were not exposed to novelty. In the (4) group-novel (GRP–NOV) condition, all three monkeys were caught and then immediately taken to a different room (animals held ~30 sec) where they were placed together in a wire mesh cage (89 × 56 × 56 cm) and then observed. The test room contained no other animals and was not familiar to the monkeys. GRP–NOV was used to evaluate the effects of exposure to novelty in animals not separated from their cagemates. The (5) isolate-novel (ISO–NOV) condition was identical to GRP–NOV, with the exception that only one monkey was taken to the test room and observed. Auditory contact was not possible between the colony room and test room. In ISO–NOV, the separated monkey was, therefore, both isolated from cagemates and exposed to novelty. In all five conditions, monkeys were observed over a 30-min period, and a blood sample was then taken for subsequent analysis of plasma cortisol levels.

The experiment was conducted during a 7-week period. Conditions were imposed at weekly intervals and each group of three monkeys was tested on a separate day. All animals were examined in the BASE condition on both week 1 and week 7. The monkeys were tested in HAN on week 3, and GRP–NOV on week 5. On weeks 2, 4, and 6, animals were tested in conditions involving separation. For each group, a different individual was tested in ISO–NOV each week. At the same time, the other two animals in the social group were examined in the SEP–HOM condition. Thus, every monkey was tested once in ISO–NOV, and twice in SEP–HOM—once with each of its two cagemates. All testing was performed between 1500 and 1700 hr.

Behavior Observations

We monitored vocalizations, locomotor activity, and manual manipulation of objects. Increases in the first two of these behaviors, and decreases in the third, have been observed to be major components of the protest reaction of infant squirrel monkeys during separation [Hennessy et al., 1979; Kaplan, 1970]. We also observed behaviors that appear to serve a scent-marking function in *Saimiri* (urine wash, rump rub, back rub, sneeze; see Hennessy et al., 1978, 1980 for complete definitions) since putative scent-marking behaviors have been reported to increase in frequency during times of excitement or tension [Latta et al., 1967; Schwartz & Rosenblum, 1980; Baldwin, 1969]. Simple frequency measures were used for all behaviors except locomotor activity. Because this behavior tends to occur in long temporal bouts, we recorded the number of 15-sec intervals in which locomotion of more than 15 cm was observed. All vocalizations of

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moderate to high intensity were scored regardless of type. Occasional vocalizations of low intensity, which did not appear indicative of distress, were not recorded. Each monkey was observed for five 2-min sessions that were separated by 4-min intervals during each test period. All observations were conducted by experienced observers who were concealed in observation booths equipped with one-way glass windows.

Blood Sampling and Cortisol Determination

Immediately following the 30-min observation period, the monkeys were rapidly anesthetized with ethyl ether. A blood sample (~0.8 ml) was then taken via cardiac puncture using a heparinized injection needle (25 gauge × 1.6 cm) and syringe. All monkeys from a group had their blood sampled within about 3–4 min of capture. The collected blood was centrifuged to separate plasma, which was then extracted and frozen until assay. Plasma cortisol was determined using a radioimmunoassay (^{3}H) kit (New England Nuclear). Pilot work showed that the chromatography step was unnecessary, and so it was omitted from the procedure.

Data Analysis

Occurrences of rump rub, back rub, urine wash, and sneeze were too infrequent for statistical purposes, so frequencies of the individual behaviors were pooled to provide a "total-scent-marking" score. For each dependent measure (vocalizations, locomotor activity, total scent marking, object manipulation, plasma cortisol), each animal's mean score on its two BASE trials and on its two SEP–HOM trials were used for comparison with scores in other conditions. There was no significant difference between the first and second BASE or SEP–HOM trials on any measure. Preliminary analysis indicated that age was not a significant factor. Comparison of the 20–22-month-old versus the 33–35-month-old subjects revealed no significant differences, with the single exception of a greater amount of vocalizing by the youngr as compared to older monkeys in the GRP–NOV condition ($p < 0.05$). Yet, even with this difference, the rank order of the frequency with which vocalizations occurred in the five conditions was identical for the younger and older groups. Therefore, to facilitate data presentation in the present report, the scores of all subjects were combined for analysis.

Nonparametric tests were used to make comparisons across conditions for vocalizations, locomotor activity, object manipulation, and total scent marking. The large number of extreme scores in particular conditions (zeros for vocalization, object manipulation, and total scent marking; 40 for locomotor activity, which was the maximum score possible in this category) made parametric tests inappropriate. For these measures, overall comparisons across conditions were made with the Friedman analysis of variance by ranks, and significant effects were further evaluated with the Wilcoxon matched-pairs signed-ranks test [Siegel, 1956]. Plasma cortisol data were analyzed with parametric analysis of variance followed by a Newman–Keuls test [Winer, 1971]. To examine the relationship between individual differences in behavioral and hormonal responses, we calculated Spearman rank correlation coefficients [Siegal, 1956] between behavioral measures and plasma cortisol levels.

RESULTS

The number of vocalizations varied greatly across conditions ($\chi_r^2 = 23.08$, $p < 0.01$; Fig. 1). Vocalizations were much more frequent in ISO–NOV than they were in any other condition (all $ps < 0.01$). With the exception of one monkey that was never observed vocalizing in the present study, every animal vocalized more in ISO–NOV than it did in any of the other four conditions. In addition, significantly more vocalizations were noted in GRP–NOV than in either the HAN or SEP–HOM conditions ($ps = 0.05$). Although specific vocalization types were not systematically differentiated, observers did

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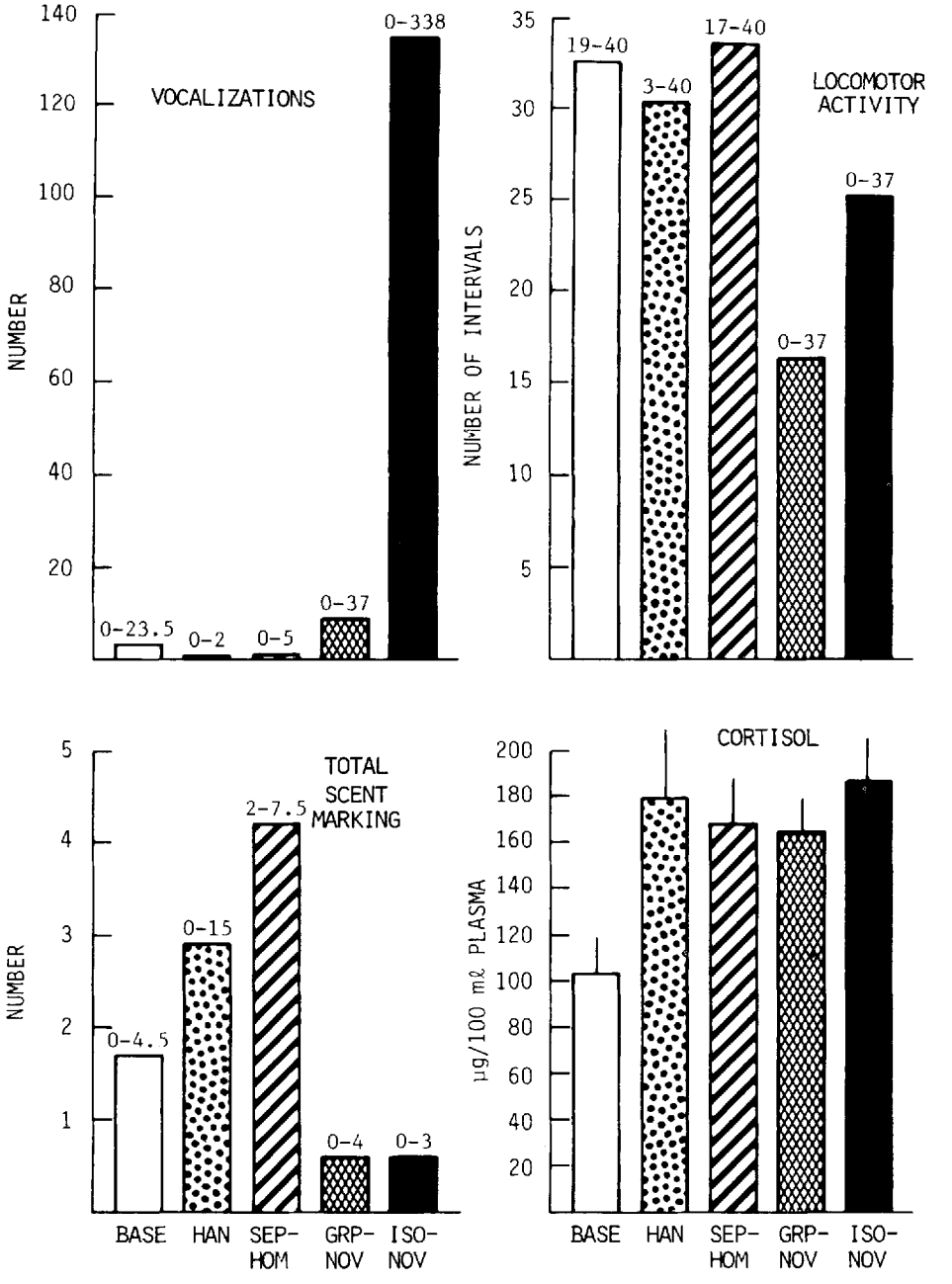


Fig. 1. Mean vocalizations, locomotor activity, and total scent marking per 10-min interval, and mean plasma cortisol levels for monkeys in each of the five conditions. Vocalizations and total scent markings are expressed as simple frequencies, and locomotor activity as the number of 15-sec intervals in which locomotion occurred. For cortisol, vertical bars represent standard errors of the mean. For vocalizations, locomotor activity, and total scent marking, which had markedly nonnormal distributions, the range of scores in each condition is presented as a measure of dispersion.

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make general notations of classifications. In ISO-NOV, 81% of all recorded utterances were high-pitched cries. These appeared to be predominantly of the type referred to as "isolation peeps" [Winter et al., 1966]. In the only other condition in which an appreciable number of vocalizations was scored, GRP-NOV, 94% were high-pitched cries, and many of these again appeared to be of the isolation peep variety. Despite the large number of high-pitched cries heard, these did not account entirely for the observed effects. An analysis of all vocalizations other than high-pitched cries was also significant ($\chi_r^2 = 13.68, p < 0.01$), and follow-up tests showed that more of these vocalizations were heard in ISO-NOV than in any other condition (all ps < 0.01).

Analysis of locomotor activity revealed a significant overall effect ($\chi_r^2 = 16.05, p < 0.01$; Fig. 1). Follow-up tests indicated that this was due to a general suppression of locomotor activity in the novel environment. That is, activity was more frequent in BASE ($p < 0.01$), HAN ($p < 0.05$), and SEP-HOM ($p < 0.01$) than it was in GRP-NOV; and more frequent in SEP-HOM than in ISO-NOV ($p < 0.05$).

Overall analysis of total scent marking was significant ($\chi_r^2 = 21.01, p < 0.01$; Fig. 1). Paired-comparisons indicated that this measure, like locomotor activity, was suppressed in novel surroundings. More scent marking occurred in HAN ($p < 0.05$) and SEP-HOM ($p < 0.01$) than in GRP-NOV; and in BASE ($p < 0.05$), HAN ($p < 0.01$) and SEP-HOM ($p < 0.01$) than in ISO-NOV. In addition, this measure showed some influence of peer separation, in that total scent marking was more often observed in SEP-HOM than in BASE ($p < 0.05$). The overall analysis of object manipulation was not significant.

Plasma cortisol values varied significantly across conditions [$F(4,44) = 5.14; p < 0.01$; Fig. 1]. A Newman-Keuls test indicated that this effect was due to all manipulations producing an equivalent elevation in plasma cortisol over baseline levels. That is, cortisol concentrations in HAN, SEP-HOM, GRP-NOV, and ISO-NOV were all greater than in BASE (all ps < 0.01), and did not differ significantly among themselves.

The correlation between cortisol and vocalizations was significant in the GRP-NOV condition ($r = +0.716, p < 0.05$). However, the relationship between these variables did not appear to be a consistent one, since no significant correlation between cortisol and vocalizations was found in any of the other four conditions. Further, no significant correlations were obtained between cortisol and either locomotor activity or total scent marking in any experimental condition.

DISCUSSION

We found that juvenile squirrel monkeys separated briefly from their peers in a novel room showed high levels of vocalization, a behavior classically used to denote protest in infants separated from their mothers. It appeared that the novel room itself was stressful for the monkeys, as indicated by an increase in vocalizations and a suppression of locomotor and scent-marking activity when all three monkeys were placed in the novel room (GRP-NOV), as compared to when all three monkeys were handled and then returned to the home cage (HAN). However, novelty alone could not account for the increase in vocalizations observed following separation in the novel room because many more vocalizations were found to occur in ISO-NOV than in the GRP-NOV condition. Similarly, the increase in vocalizations could not be accounted for solely by the removal of a peer since vocalizations were much more frequent in ISO-NOV than they were in SEP-HOM. Rather, it appears that the monkey must not only be separated from a peer, but it must, in addition, be totally isolated and/or exposed to novel surroundings for a substantial increase in vocalizations to occur.

It is of interest that novelty inhibited locomotor activity in our isolated juveniles, since enhanced locomotion is a component of protest in infant *Saimiri* and occurs during separation in novel surroundings [Hennessy & Kaplan, in press; Jones & Clark, 1973]. These contrasting findings might result from some ontogenetic factor, for instance, an age-re-

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lated change in responsiveness to novelty. They might also reflect an artifact inherent in the infant separation paradigm; that is, the infant is separated from a figure (mother or surrogate) that typically elicits clinging, a response incompatible with locomotor activity.

The only sign of behavioral disturbance found in the SEP-HOM condition was an increase in total scent marking relative to that observed in BASE. This effect should, in a strict sense, be attributed to removal of a peer combined with handling since both of these manipulations were imposed in SEP-HOM, and when monkeys were handled only (HAN), they evinced scent-marking levels that were not significantly different from those in either BASE or SEP-HOM. The severe suppression of scent-marking activity induced by the novel room may have obscured any influence on this behavior of peer removal or handling in the ISO-NOV and GRP-NOV conditions.

Plasma cortisol levels were significantly elevated by the handling procedure as seen in the comparison of the HAN and BASE conditions. No additional effect of peer separation or exposure to novelty on cortisol levels was observed. Some caution in interpretation is suggested here, however, since it is possible that the substantial response to handling alone may have masked effects of the additional manipulations. Nevertheless, our results contrast sharply with those for mother and infant squirrel monkeys, which show large increases in plasma cortisol 30 min following separation, but little or no response to disturbance control conditions at least as traumatic as involved in the HAN condition of the present study [Mendoza et al., 1978; Coe et al., 1978].

Overall, the present study demonstrates that brief disruption of relationships with peers can produce behavioral signs of distress in juvenile squirrel monkeys. This effect was clear only when the juvenile was isolated in a novel room and not when it remained in the home cage in the presence of another social partner during separation. This pattern is similar to that seen in infant rhesus, which show greater protest when isolated in novel surroundings than they do when left in the home cage with familiar animals during maternal separation [Hinde & Davis, 1972]. In contrast to behavior, pituitary-adrenal activity was not found to be affected by separation from peers under the conditions of the present study. A similar lack of cortisol response has also been obtained recently in a study of peer separation with somewhat younger squirrel monkeys [Vogt, personal communication]. In terms of pituitary-adrenal effects of separation, the juveniles more closely resembled surrogate-reared infants, which showed no effect of surrogate separation on cortisol levels at 30 min [Hennessy et al., 1979], than they did mother-reared infants or their mothers, which showed large cortisol responses to brief separation [Coe et al., 1978; Mendoza et al., 1978]. Perhaps a social relationship must be particularly intense, as is that between mother and infant, for brief separation to be sufficient stimulus, in and of itself, to activate the pituitary-adrenal system.

CONCLUSIONS

1) Juvenile female squirrel monkeys exhibited behavioral disturbance under conditions of brief separation from peers.

2) The behavioral effects varied greatly with the separation procedure used.

3) Plasma cortisol levels were elevated following various procedures involving a brief period of handling, but under the conditions of the present study, did not show any effect of peer separation.

4) The behavioral (vocalization), but not cortisol, findings were similar to those that have been obtained with maternally-separated infants.

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